

Advanced Cognitive Neuroscience

Week 38: What is it all about?
September 14th 2026

Academic regulations - objectives

KNOWLEDGE

- describe the anatomy and physiology of the human brain, and explain the brain basis of cognitive function
- contrast different cognitive neuroscience methods in terms of their strengths and weaknesses, and use this knowledge to develop appropriate experimental research for investigating different cognitive functions of the brain.

Academic regulations - objectives

SKILLS

- run experiments using neuroimaging and/or neurophysiological measurement equipment
- use advanced statistical methods to make inferences about cognitive brain functions from neuroimaging and/or neurophysiological data.

Academic regulations - objectives

COMPETENCES

- independently identify the appropriate measurement technology and experimental designs for investigating different cognitive functions
- identify cases in which statistical methods taught in the course can be applied to domains outside of cognitive neuroscience.

Overall idea of the course

- We will focus on magnetoencephalography (MEG)
 - but leverage magnetic resonance imaging to do source reconstruction of the MEG signals
- We will investigate how subjective experience is build in the brain
- We will learn how to apply multivariate statistics to reduce these very rich data sets (MEG: 306,000 data points per second)
- We will learn how to facilitate the rich spatio-temporal data that MEG provides
- We will learn about the brain while doing so

Us

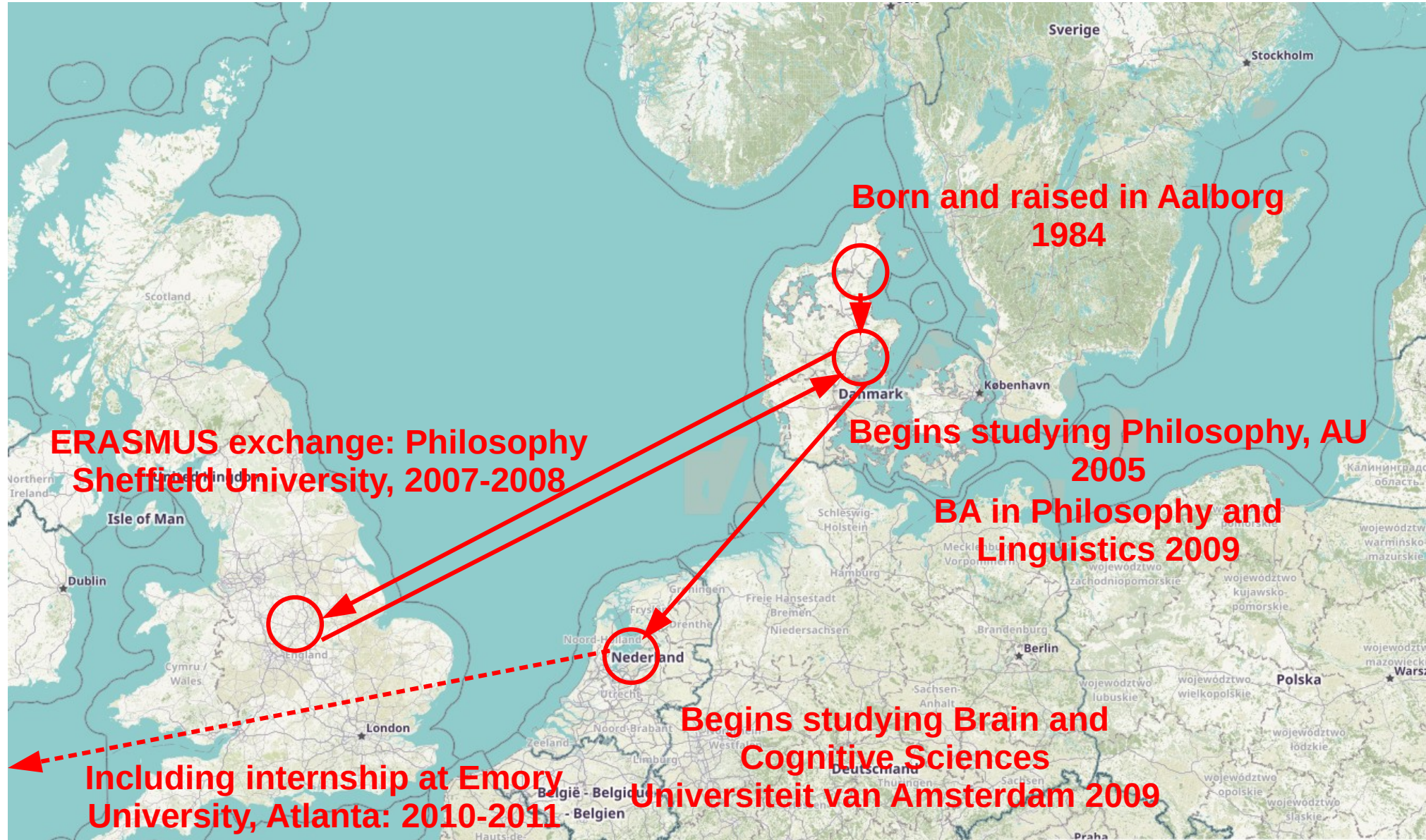


Laura Bock Paulsen
<https://github.com/laurabpaulsen>



Lau Møller Andersen
laumollerandersen.org

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**Born and raised in Aalborg
1984**

**ERASMUS exchange: Philosophy
Sheffield University, 2007-2008**

**Begins studying Philosophy, AU
2005**

**BA in Philosophy and
Linguistics 2009**

**Begins studying Brain and
Cognitive Sciences
Universiteit van Amsterdam 2009**

**Including internship at Emory
University, Atlanta: 2010-2011**

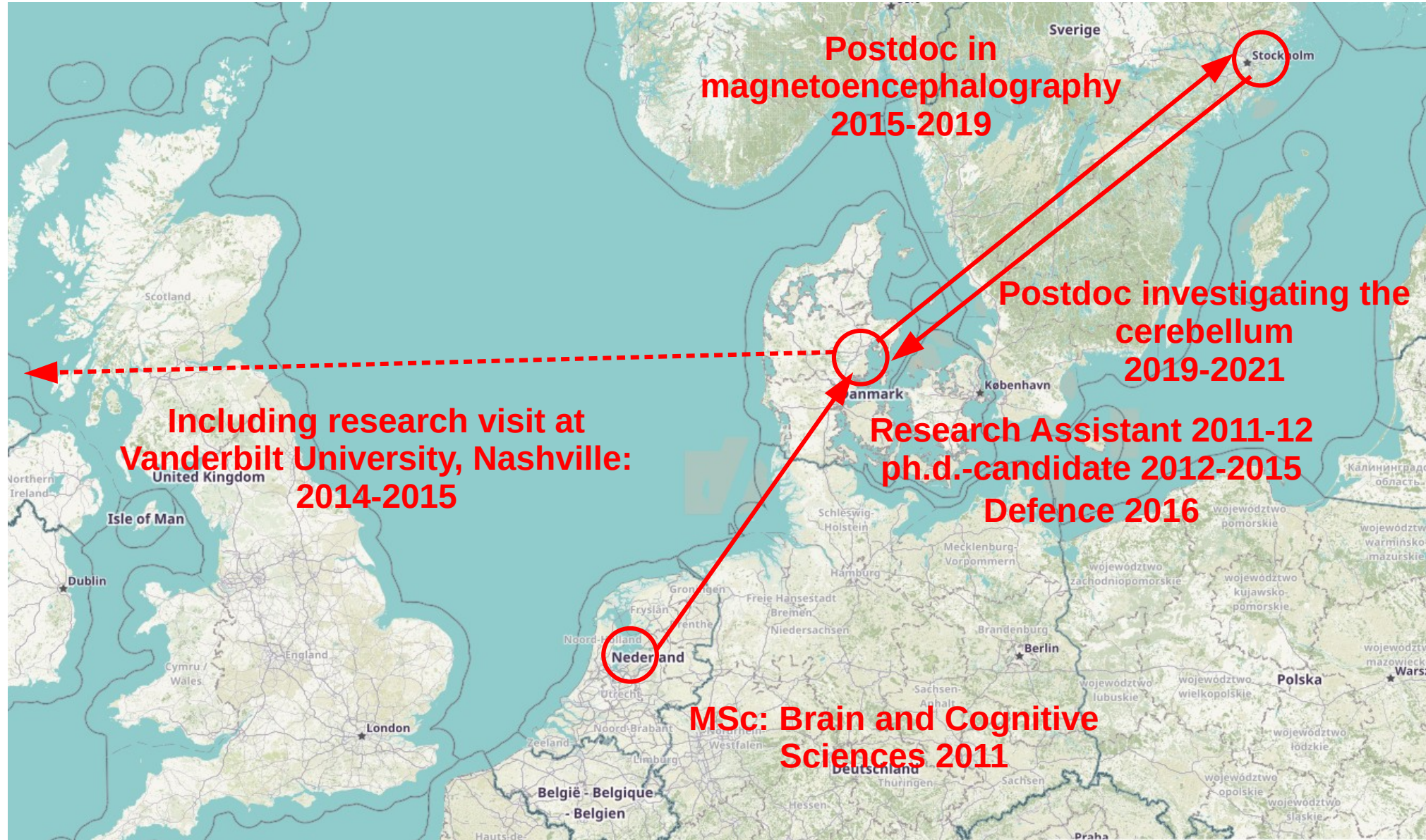
**Postdoc in
magnetoencephalography
2015-2019**

**Postdoc investigating the
cerebellum
2019-2021**

**Including research visit at
Vanderbilt University, Nashville:
2014-2015**

**Research Assistant 2011-12
ph.d.-candidate 2012-2015
Defence 2016**

**MSc: Brain and Cognitive
Sciences 2011**



2021-2024

Welcome to AIAS

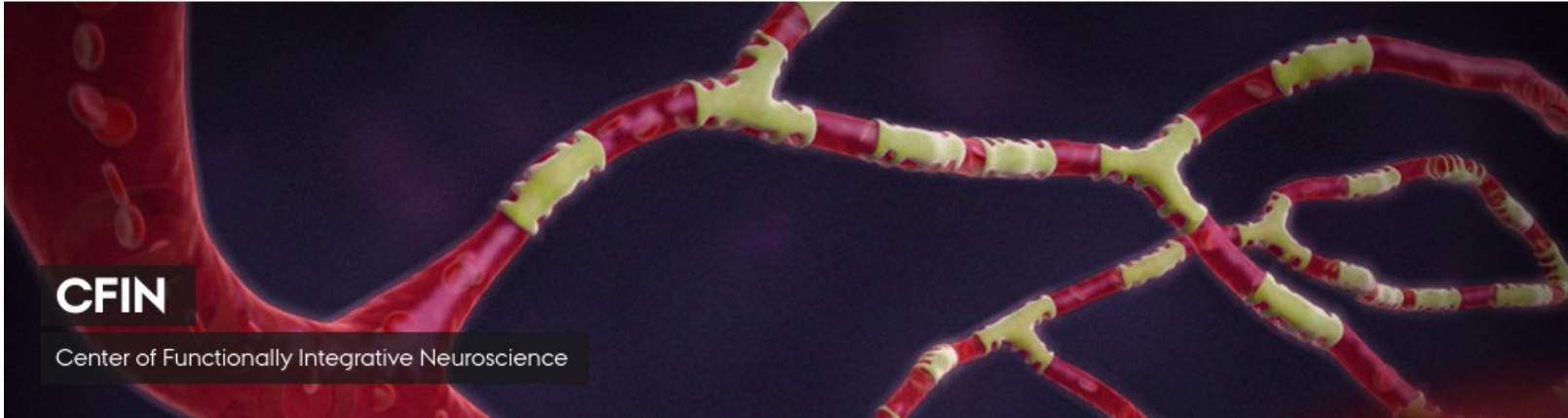
Depth of Research, Crossing Boundaries

A new approach to interdisciplinary solutions



CFIN

Center of Functionally Integrative Neuroscience

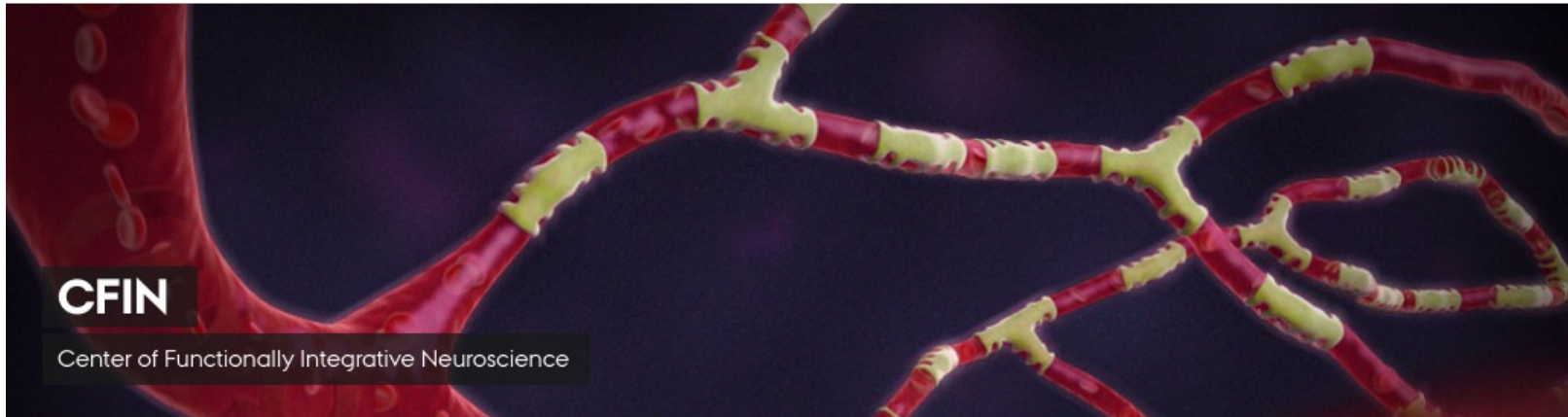


2024 – onwards

Main affiliation



Secondary affiliation



My current interest

Review

Can EEG and MEG detect signals from the human cerebellum?

[Lau M. Andersen](#)^{a b}  , [Karim Jerbi](#)^{c d}, [Sarang S. Dalal](#)^a

The cerebellar clock: Predicting and timing somatosensory touch

[Lau M. Andersen](#)^{a b c}  , [Sarang S. Dalal](#)^a

Somatosensory responses to nothing: An MEG study of expectations during omission of tactile stimulations

[Lau M. Andersen](#)  , [Daniel Lundqvist](#)

Research Article | Research Article: New Research, Cognition and Behavior

Detection of Threshold-Level Stimuli Modulated by Temporal Predictions of the Cerebellum

[Lau M. Andersen](#) and [Sarang S. Dalal](#)

Review

The role of the cerebellum in timing

[Lau M Andersen](#)^{1 2 3} , [Sarang S Dalal](#)¹ 

New Results

 [Follow this preprint](#)

Somatosensory timing and cerebellar-basal ganglia beta-band interactions in Parkinson's disease

 Victor Pando-Naude,  Lau Møller Andersen

You

- My aim:
 - Every time I address you, I will do it with your name
 - Some times I will be wrong, and that might cause some humiliation both ways
 - Please correct me when I am wrong

Me as a lecturer

Me asking questions

- I will every now and then pick a student at random to answer questions
- I want to emphasise that I am only picking students that I judge are paying attention
 - i.e.: I am not trying to humiliate students not paying attention
- It is completely okay to say “I do not have an answer”; then I’ll just pass it on to someone else

You asking questions

- Does such a thing as a stupid question exist?
- I don't care if they do or not
- You ask a question because you want to know ...
- ... and I answer because I want to tell you

Exam; portfolio; deadline 15-12-2025

SEE SYLLABUS FOR DETAILS

1. *Video explainer to your peers:* **Individual assignment (25 % of grade):**
2. *Report on MEG acquisition and MEG analysis:* **Project group assignment (50 % of grade):**
3. *Oral presentation on set topic:* **Project group assignment (25 % of grade):**

Re-examination

Will be done during January (tip: stay on schedule, such that you can enjoy Christmas)

The structure of lectures

- Mondays 10-12 (NB: Academic quarter)
- A mix between the general theoretical framework and formulas
- Code will be incorporated
- Group discussions
- Please don't hold back on your questions

The structure of classes

- Wednesdays 13-15 (NB: Academic quarter)
 - In classes, you will generally be implementing code and adapting it to your analyses that go into your report
 - Will be led by Laura Bock Paulsen

1. *Video explainer to your peers: Individual assignment (25 % of grade):*

- Task: Make a video explainer of maximally 5 minutes; this should explain the underlying processes that give rise to a measurable magnetic field outside the head. The following concepts should be included: *post-synaptic potentials, current dipole and equivalent current dipole, open field vs closed field, radial and tangential sources, volume conduction and evoked responses*. In general the level of the video explainer should be aimed at your peers and the relevant materials can be found in Chapters 1-3 of the textbook. You are not allowed to use props or slides in the background.
- Evaluation criteria: to what degree are the concepts listed above. **clearly, succinctly** and **correctly** described. It will count negatively if content is read aloud
- Deadline for receiving feedback: Week 41 (Tuesday 23.59)

From AU's centre of education

Regarding editing options, AU's video platform Panopto offers a bit of editing of the material, but it is very limited. I would recommend that you download **Blackmagic's DaVinci Resolve**, which you can get for free. It has reached a fairly high level within the industry and provides many tools even in the free version. It's easy to use if you already have some editing experience; otherwise, there are plenty of great tutorials on YouTube that can help you get started.

Uploading video for exam

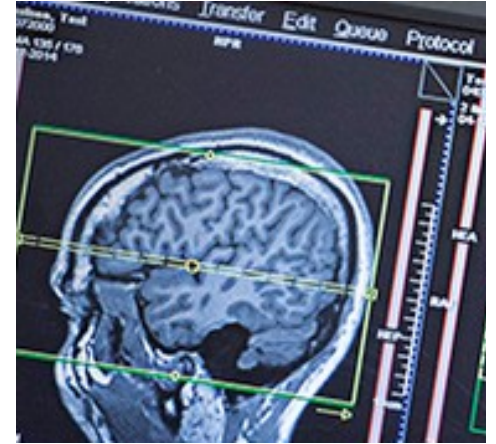
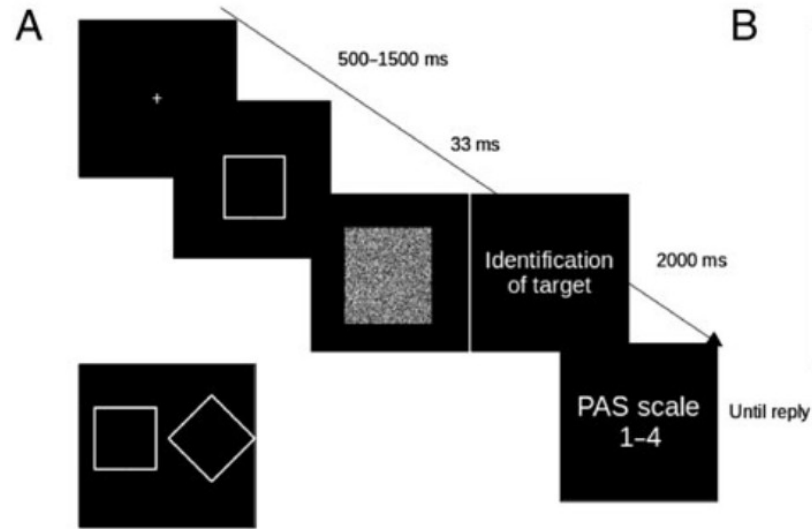
- You may also provide a link for the video
 - Make sure it is stable!

Workshop

SUBJECTIVE EXPERIENCE



from Center of Functionally
Integrative Neuroscience



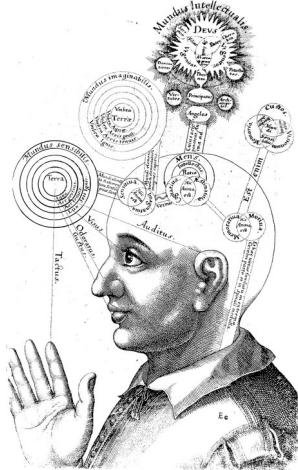
Andersen LM, Pedersen MN, Sandberg K, Overgaard M (2016)
Occipital MEG Activity in the Early Time Range (<300 ms)
Predicts Graded Changes in Perceptual Consciousness.
Cerebral Cortex 26:2677–2688.
<https://doi.org/10.1093/cercor/bhv108>

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Workshop

Table 1 The Perceptual Awareness Scale (PAS)

Label	Description [from Ramsøy and Overgaard (2004)]
(1) No Experience (NE)	No impression of the stimulus. All answers are seen as mere guesses.
(2) Weak Glimpse (WG)	A feeling that something has been shown. Not characterized by any content, and this cannot be specified any further.
(3) Almost Clear Experience (ACE)	Ambiguous experience of the stimulus. Some stimulus aspects are experienced more vividly than others. A feeling of almost being certain about one's answer.
(4) Clear Experience (CE)	Non-ambiguous experience of the stimulus. No doubt in one's answer.



Andersen LM, Pedersen MN, Sandberg K, Overgaard M (2016) Occipital MEG Activity in the Early Time Range (<300 ms) Predicts Graded Changes in Perceptual Consciousness. *Cerebral Cortex* 26:2677–2688.
<https://doi.org/10.1093/cercor/bhv108>

Ramsøy TZ, Overgaard M (2004) Introspection and subliminal perception. *Phenomenology and the Cognitive Sciences* 3:1–23.
<https://doi.org/10.1023/B:PHEN.0000041900.30172.e8>

```
##% RUN EXPERIMENT
```

```
## initialise
experiment = Experiment()

## setting up experiment
experiment.open_GUI()
experiment.set_experiment_parameters()
experiment.check_user()
experiment.write_to_terminal('refresh_rate')
experiment.define_io_files()
experiment.write_to_terminal('setting_path')
experiment.define_texts()
experiment.create_experiment_window()
experiment.define_visual_stimuli()
experiment.present_instructions('welcome')

## practice
experiment.present_instructions('practice')
experiment.run_practice()

## experiment
experiment.set_experiment_parameters()
experiment.present_instructions('experiment')
experiment.run_experiment()

## thank you
experiment.present_instructions('thank_you')
```

Check out the experiment on your own computer

https://github.com/ualsbombe/2026_advanced_cognitive_neuroscience/tree/main/week37/MEG_experiment



Behavioural data

```
[1] "study_group_1_2026_09_08_144940_experiment_data.csv"
```

```
1 2 3 4
248 250 68 34
```

```
[1] "study_group_4_2026_09_08_114022_experiment_data.csv"
```

```
1 2 3 4
136 284 128 52
```

```
[1] "study_group_7_2026_09_09_105733_experiment_data.csv"
```

```
1 2 3 4
132 320 117 31
```

```
[1] "study_group_8_2026_09_09_131040_experiment_data.csv"
```

```
1 2 3 4
77 256 184 83
```

```
[1] "0192_2026_09_09_154435_experiment_data.csv"
```

```
1 2 3 4
178 249 111 62
```

```
[1] "0193_2026_09_10_101608_experiment_data.csv"
```

```
1 2 3 4
90 248 175 87
```

```
[1] "0194_2026_09_10_131832_experiment_data.csv"
```

```
1 2 3 4
64 129 96 311
```

```
[1] "0195_2026_09_10_160615_experiment_data.csv"
```

```
1 2 3 4
297 197 76 30
```



Behavioural data



MEG data

Wednesday; only the subject

MR		
	Wednesday 16	Group number
	09.00-09.20	5
	09.20-09.40	4
	09.40-10.00	6
	10.00-10.20	7
	10.20-10.40	1
	10.40-11.00	2
	11.00-11.20	3
	11.20-11.40	8

2. *Report on MEG acquisition and MEG analysis: Project group assignment (50 % of grade):*

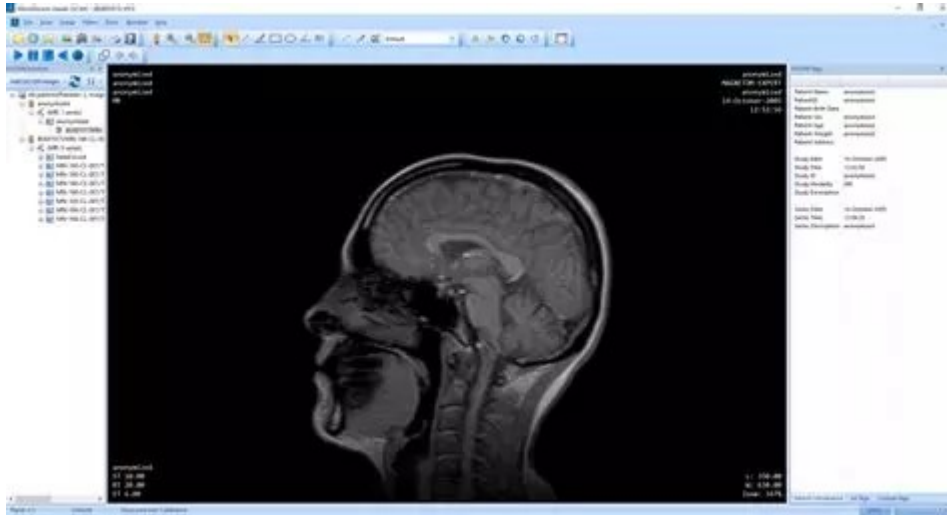
- Task: The report must include an **introduction**, based on relevant literature; operationalisable **hypotheses**; a clear **methods** section, which must describe the experiment and the data acquisition procedure; a succinct **results** section with both behavioural data and MEG data; a **discussion** addressing limitations and the suitability of the method for investigating subjective experience. Data from all 8 participants **must** be used and **at least one** analysis **must** be based on multivariate statistics. A maximum of 6 pages per co-author. Code as appendix.
- Evaluation Criteria: to what degree has **relevant** literature been chosen for the introduction; to what degree has **operationalisable** hypotheses been formulated; to what degree has the methods section **clearly** described the experiment and the data acquisition procedure; to what degree are the results **succinctly** described; to what degree are the limitations and the suitability of the method **relevantly** described.
(**Remember** to use all participants and to use a multivariate analysis)
(**Also remember**: novelty is not a requirement: you can use the hypotheses from my old paper; they still have be argued though:
Andersen, L.M., Pedersen, M.N., Sandberg, K., Overgaard, M., 2016.
Occipital MEG Activity in the Early Time Range (<300 ms) Predicts Graded Changes in Perceptual Consciousness. Cerebral Cortex 26, 2677–2688.
<https://doi.org/10.1093/cercor/bhv108>)
- Deadline for receiving feedback: Week 46 (Friday 23.59)

Study groups

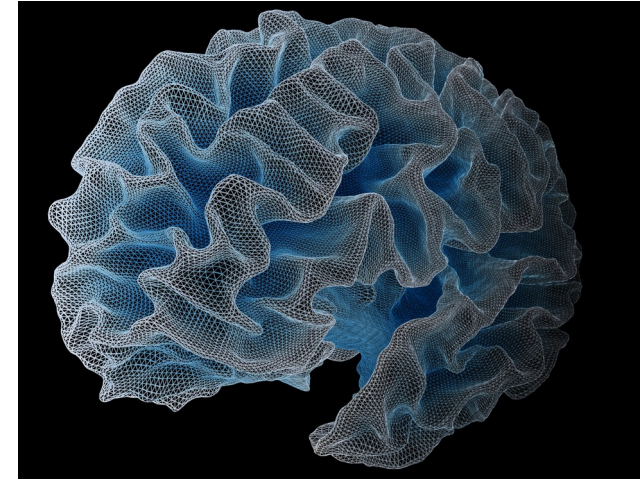
Study group no	Full name
1	Denitsa Delyanova Pehlivanova
1	Akaran Sivakumar
1	Anne Christine Vig Jensen
1	Christian Vejen Barnkob
1	Valeria Alladio
2	Vyara Krasteva
2	Alexander Juby-Rasmussen
2	Daniel Lindblad Sørensen
2	Nadia Kinga Wójtowicz
3	Anders Herzog Varan
3	Andreas Møldrup Holst
3	Clara Holst
3	Zuzanna Katarzyna Zyla
3	Cosma Carlotta Rost
4	Gustav Idun Sloth
4	Halfdan Nordahl Fundal
4	Ida Elmoose Brøcker
4	Paula Samide
4	Lilla Fürtön

5	Johannes Eide Rambøll
5	Laura Emilie Lundbye
5	Louie Mølgaard Hessellund
5	Rose Lina Reinhold
5	Reka Forgo
6	Frederikke Lykke Korreborg
6	Liv Drasbek
6	Lucas Johannes Simonsen
6	Lydia Pauli Rambrand
6	Anne Skamris Holm
7	Marie Vestergaard Thomsen
7	Mikkel Lindhardt Glavind Andersen
7	Nanna Klitgaard Damgaard
7	Niels Engedal
8	Sara Eline Røeder Nielsen
8	Sissel Højgaard Vang-Pedersen
8	Vilma Bejer Kristiansen
8	Julianne Silke Swärd

FreeSurfer



<https://lasopavegan481.weebly.com/blog/best-dicom-viewer-for-windows-10>



From FreeSurfer

<https://surfer.nmr.mgh.harvard.edu/>

Presentation topics

- 1) The visual system, as understood through MEG
- 2) The auditory system, as understood through MEG
- 3) The somatosensory system, as understood through MEG
- 4) Cognitive components and change detection
- 5) Links between respiration, behaviour and MEG
- 6) The motor system, as understood through MEG
- 7) Language comprehension, as understood through MEG
- 8) Subjective experience, as understood through MEG

Choosing them

- Getting a random seed from random.org (between 0 and 255)

```
import random

random.seed('from random.org')

k = 8

population = range(1, k+1)
samples = random.sample(population, k=k)

print(samples)
```

■ Evaluation Criteria:

Criteria	Specifics	Percentage
Clear agenda	The structure is clear throughout; e.g. <i>method</i> , <i>results</i> and <i>interpretation</i> are clearly separated	10
Time management	30 min for the presentation; 15 min for Q&A. Make sure to allot time for each part of the chapter	5
Understanding of MEG fundamentals - e.g. basic physics, sensor types, source reconstruction	Make sure to use the right terminology and explain succinctly what terms mean	20
Clear presentation of summary text	Extract core concepts from the chapter and explain core figures	0-40

Clear appraisal of experiment text	Summarise the study, explain the experimental design, explain results, the interpretation of the results and the limitations of the study	0-40
Visual design	Coherent design, legible slides and no unnecessary information	5
Presentation skills	No reading aloud, control of the material	10
Handling of Q&A	How do you engage with the questions of fellow students? e.g. do you get the question, can you answer it and so on?	10

Study counselling

Who are we?



Christian: Cognitive
Science



Kaysa: Linguistics



Iben: Nordic
languages and
literature

[Link for website](#)

The course plan

Week 37:

Monday: Workshop paradigm: Measuring visual subjective experience + MR Recordings (Lau)

Tuesday-Thursday: Workshop 0: MEG: Measuring and predicting visual subjective experience

Friday: Intro to coding pipeline (Laura)

The course plan

Week 38:

Monday: What is it all about? (Lau)

Workshop 1: MR-recordings (Laura & Lau)

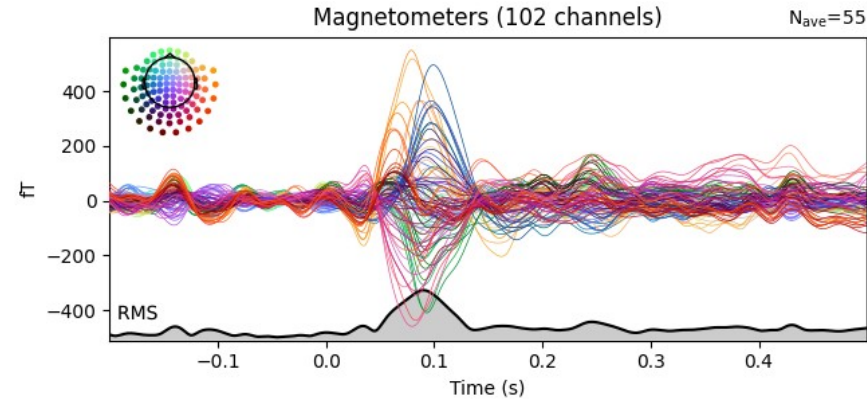
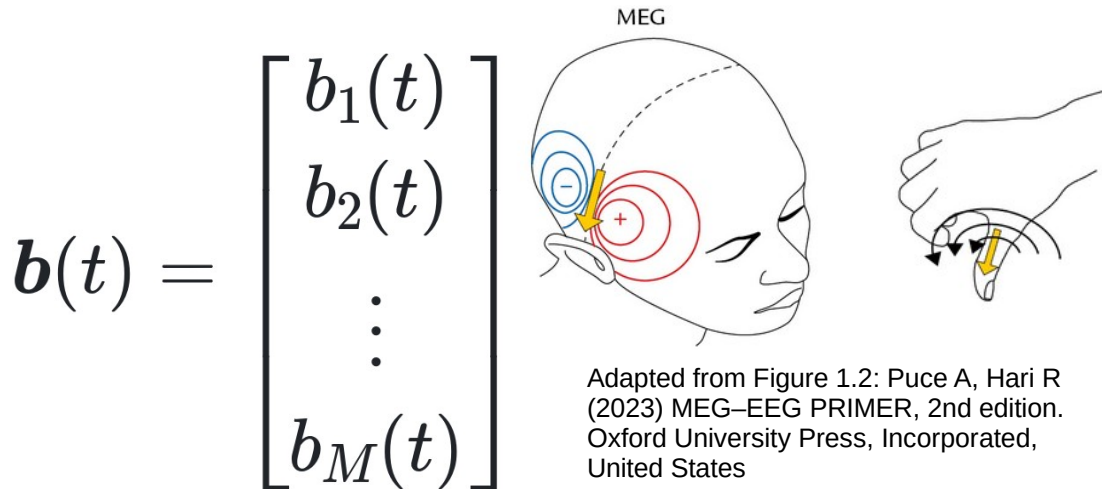
The course plan

Week 39:

Monday: Basic physiology and Evoked responses (Lau)

Wednesday: Preprocessing and evoked responses (Laura)

Readings: Chapters 3, 10 & 12, Puce A, Hari R (2023)



https://mne.tools/stable/auto_tutorials/evoked/20_visualize_evoked.html#plotting-signal-traces

The course plan

Week 40:

Monday: Forward modelling and inverse modelling (Lau)

Wednesday: Source reconstruction (Laura)

Readings:

FieldTrip video: <https://www.youtube.com/watch?v=3Q8HLHNieuI>

Chapters 1-3 Sekihara K., Nagarajan S (2008)

$$\mathbf{s}(\mathbf{r}, t) = \begin{bmatrix} s_x(\mathbf{r}, t) \\ s_y(\mathbf{r}, t) \\ s_z(\mathbf{r}, t) \end{bmatrix} \quad \mathbf{L}(\mathbf{r}) = [l_x(\mathbf{r}), l_y(\mathbf{r}), l_z(\mathbf{r})]$$
$$\mathbf{b}(t) = \int_{\Omega} \mathbf{L}(\mathbf{r}) \mathbf{s}(\mathbf{r}, t) d^3r$$

The course plan

Week 40:

Monday: Forward modelling and inverse modelling (Lau)

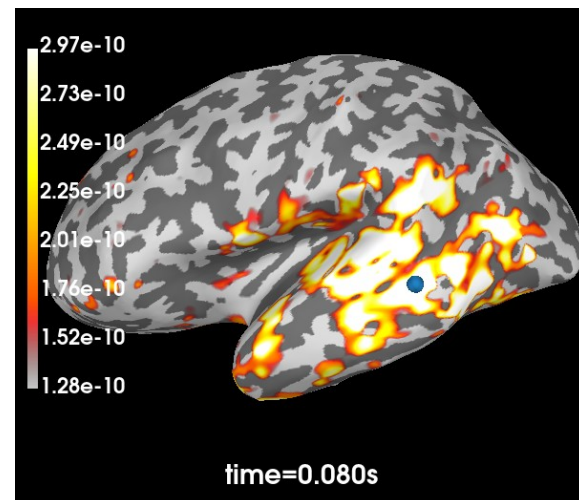
Wednesday: Source reconstruction (Laura)

Readings:

FieldTrip video: <https://www.youtube.com/watch?v=3Q8HLHNieuI>

Chapters 1-3 Sekihara K., Nagarajan S (2008)

$$\hat{\mathbf{v}}_{vox}(t) = \mathbf{L}_V^T (\mathbf{G} + \epsilon \mathbf{I})^{-1} \mathbf{b}(t)$$



The course plan

Week 41:

Monday: Multivariate statistics (Laura) (Lecture)

Wednesday: Multivariate statistics (Laura) (Coding)

Readings:

King J-R, Dehaene S (2014) Characterizing the dynamics of mental representations: the temporal generalization method.

Trends in Cognitive Sciences 18:203-210.

<https://doi.org/10.1016/j.tics.2014.01.002>

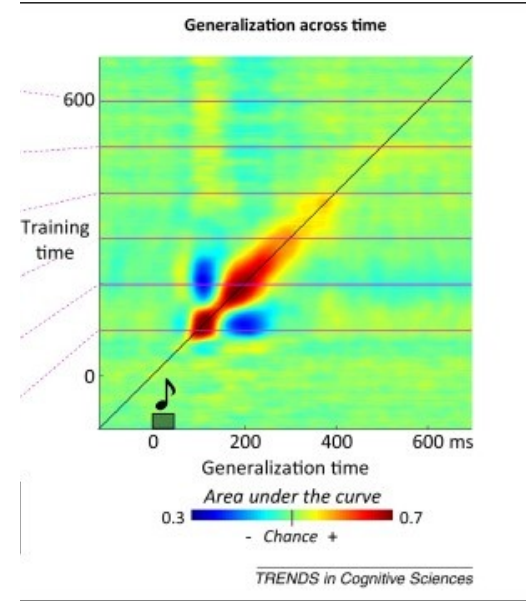
Hebart MN, Baker CI.

Deconstructing multivariate decoding for the study of brain function.

Neuroimage. 2018 Oct 15;180(Pt A):4-18.

doi: 10.1016/j.neuroimage.2017.08.005. Epub 2017 Aug 4.

PMID: 28782682; PMCID: PMC5797513



The course plan

Week 42:

Autumn Break

Readings: Go dig up potatoes instead!

The course plan

Mid-term evaluation

The course plan

Week 43:

Monday: Inverse modelling: beamforming (Lau) (Lecture)

Wednesday: Inverse modelling: beamforming (Laura) (Coding)

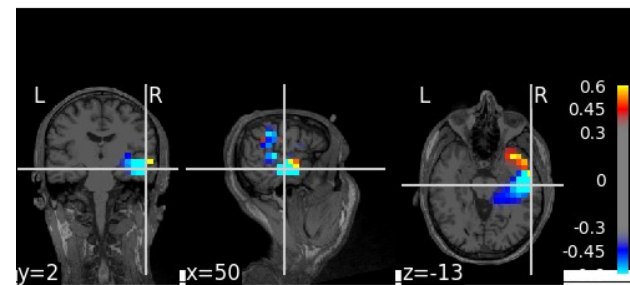
Readings:

FieldTrip video: https://www.youtube.com/watch?v=pE0WAKd_Ve4

From a signal perspective: <https://www.youtube.com/watch?v=A1n5Hhwtz78>

Chapter 4 Sekihara K., Nagarajan S (2008)

$$\hat{s}(\mathbf{r}, t) = \frac{\mathbf{l}^T(\mathbf{r})\mathbf{R}^{-1}\mathbf{b}(t)}{[\mathbf{l}^T(\mathbf{r})\mathbf{R}^{-1}\mathbf{l}(\mathbf{r})]}$$



https://mne.tools/stable/auto_tutorials/inverse/50_beamformer_lcmv.html#on-mri-slices-orthoview-2d

Week 44:

Monday From single subject analysis to group analysis (Laura)

Wednesday: Work on your reports (Laura)

Readings: Monday: From single subject analysis to group analysis

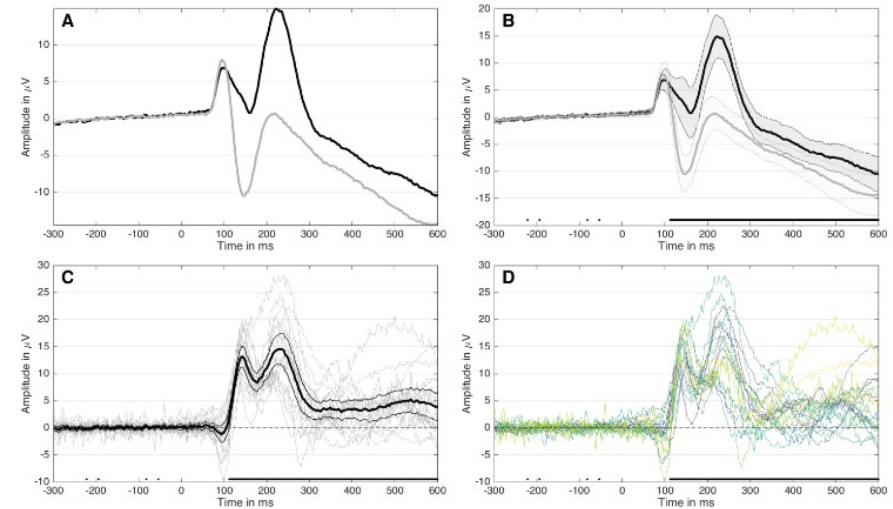
Rousselet, G.A., Foxe, J.J. and Bolam, J.P. (2016),

A few simple steps to improve the description of group results in neuroscience.

Eur J Neurosci, 44: 2647-2651. <https://doi.org/10.1111/ejn.13400>

Video: https://www.youtube.com/watch?v=PmBeGKqMJ_Y

Video: <https://www.youtube.com/watch?v=cx2YavOwD8w>



The course plan

Week 45:

Monday: What about that other cortex? - the cerebellar one (Lau)

Wednesday: Work on your reports, continued (Laura)

Readings:

Sokolov AA, Miall RC, Ivry RB (2017) The Cerebellum:

Adaptive Prediction for Movement and Cognition.

Trends in Cognitive Sciences 21:313–332. <https://doi.org/10.1016/j.tics.2017.02.005>

Andersen LM, Dalal SS (2024) The role of the cerebellum in timing.

Current Opinion in Behavioral Sciences 59:101427.

<https://doi.org/10.1016/j.cobeha.2024.101427>

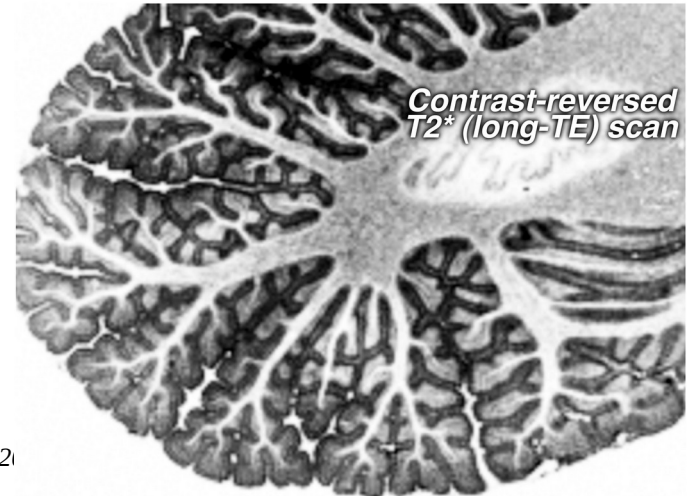
RESEARCH ARTICLE | BIOPHYSICS AND COMPUTATIONAL BIOLOGY | 



The human cerebellum has almost 80% of the surface area of the neocortex

Martin I. Sereno , Jörn Diedrichsen , Mohamed Tachrount, Guilherme Testa-Silva, Helen d'Arceuil, and Chris De

Zeeuw  [Authors Info & Affiliations](#)



The course plan

Week 46:

Monday: Optically pumped magnetoencephalography (Laura)

Wednesday: Respiration (Laura)

Readings: Monday:

Boto E, Holmes N, Leggett J, et al (2018)

Moving magnetoencephalography towards real-world applications with a wearable system.

Nature. <https://doi.org/10.1038/nature26147>

Video: <https://spectrum.ieee.org/a-new-wearable-brain-scanner>

Tierney TM, Levy A, Barry DN, et al (2021)

Mouth magnetoencephalography: A unique perspective

on the human hippocampus. *NeuroImage* 225:117443.

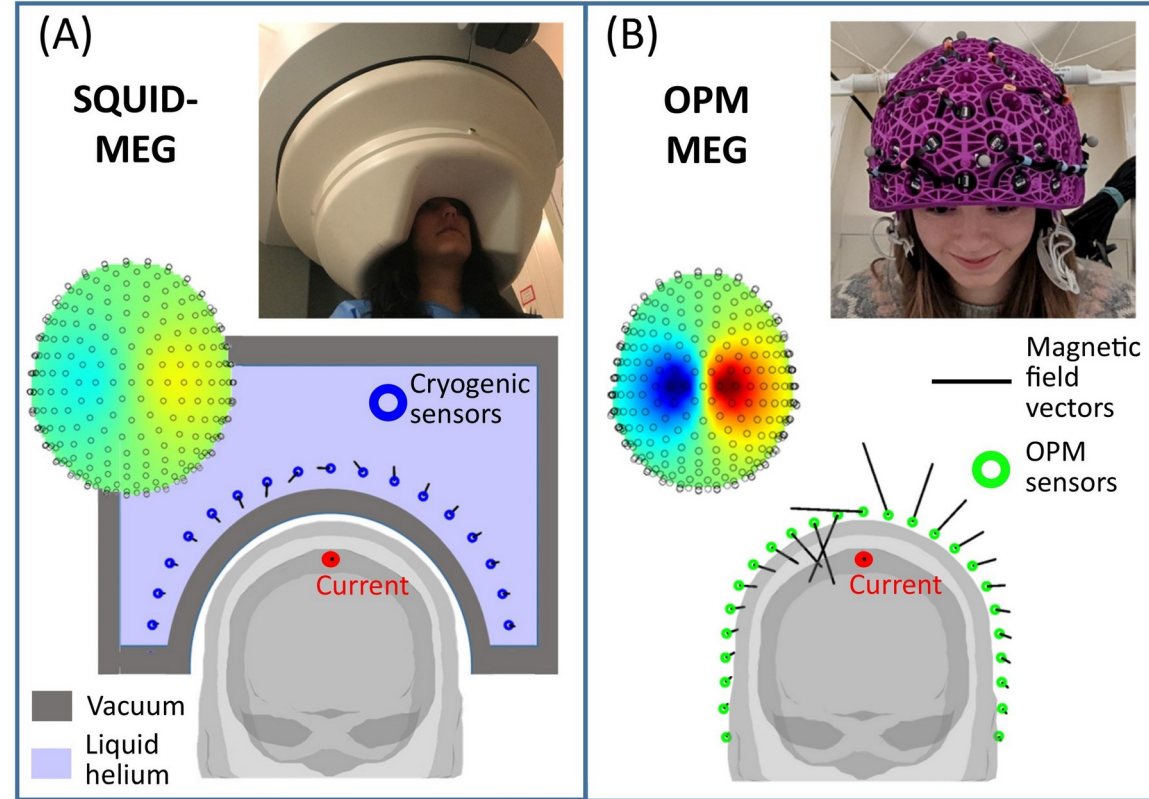
<https://doi.org/10.1016/j.neuroimage.2020.117443>

Wednesday: Heck DH, Correia BL, Fox MB, et al (2022)

Recent insights into respiratory modulation of brain activity offer new perspectives

t *Biological Psychology* 170:108316. <https://doi.org/10.1016/j.biopsycho.2022.108316>

The course plan



Trends in Neurosciences

Brookes MJ, Leggett J, Rea M, et al (2022) Magnetoencephalography with optically pumped magnetometers (OPM-MEG): the next generation of functional neuroimaging. Trends in Neurosciences 45:621–634. <https://doi.org/10.1016/j.tins.2022.05.008>

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OPMs in Aarhus



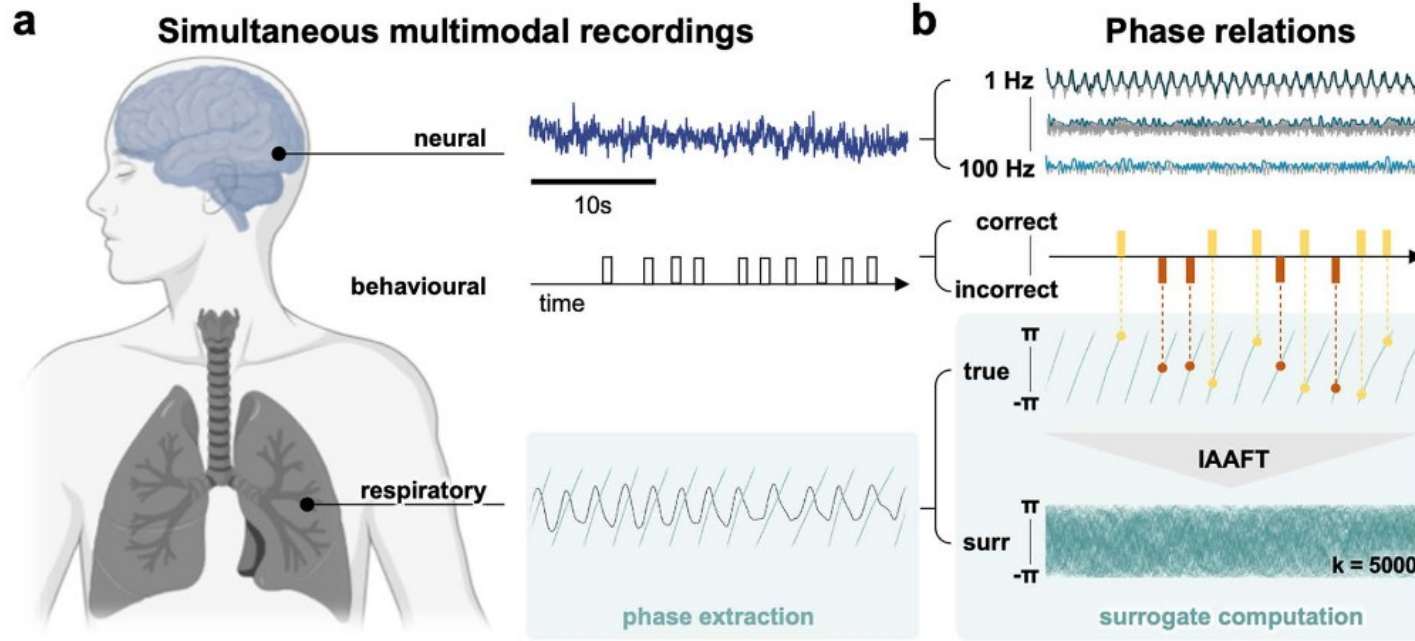
OPMs in Aarhus



OPMs in Aarhus



Respiration



Berther, T., Balestrieri, E., Saltafossi, M., Paulsen, L.B., Andersen, L.M., Kluger, D.S., 2026. Robust circular cluster-based statistics for respiration-brain coupling. PLOS Computational Biology 22, e1014672. <https://doi.org/10.1371/journal.pcbi.1014672>

Week 47:

No teaching: prepare presentations!

Readings: None

Week 48:

Monday: Oral presentations (part 1) (Lau)

Wednesday: Oral presentations (part 2) (Lau)

Readings: Student presentations: (see below)

Week 49:

Monday: Oral presentations (part 3) (Lau)

Wednesday: Oral presentations (part 4) (Lau)

Readings: Student presentations: (see below)

The course plan

Final evaluation

The course plan

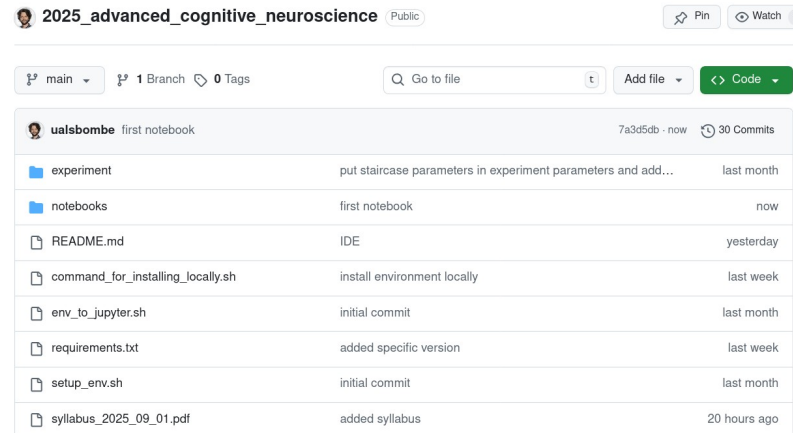
Week 50:

Wednesday: What was it all about (Lau)

Readings: Baillet S (2017) Magnetoencephalography for brain electrophysiology and imaging. Nat Neurosci 20:327–339. <https://doi.org/10.1038/nn.4504>

Keeping up to date - GitHub

- https://github.com/ualsbombe/2026_advanced_cognitive_neuroscience
- Slides will be made available here
- Notebooks will be made available here
 - upload to your member files on UCloud



Basic physiology

Chapters 1-2 & 4, Puce A, Hari R (2023)

Week 39:

Monday: Basic physiology and Evoked responses (Lau)

Wednesday: Preprocessing and evoked responses (Laura)

Readings: Chapters 3, 10 & 12, Puce A, Hari R (2023)